

Seongkwang Kim

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Professional Experience

- Assistant Professor, School of Cybersecurity, Korea University, Seoul, Korea Mar. 2026 - Present
- Senior Engineer in Samsung SDS, Seoul, Korea Mar. 2022 - Feb. 2026
 - Research on MPCitH/VOLeith-based signature
 - * Main contributor in AIMer team (NIST R1 candidate, KpqC selected)
 - * Vector semi-commitment applied to MPCitH/VOLeith-based signature schemes
 - Research on authenticated encryption with associated data (AEAD)
 - * Design and security proof of efficient beyond-birthday-bound AEAD
 - * Design and security proof of nonce-misuse resistant AEAD
 - Research on privacy-preserving protocols
 - * Privacy-preserving record linkage using circuit-based private set intersection
 - * Oblivious pseudorandom function based on oblivious key-value store

Education

- Ph.D. in Information Security Mar. 2018 - Feb. 2022
 - Where: KAIST, Daejeon, Korea
 - Advisor: Jooyoung Lee
 - Research Area: HE-friendly ciphers, transciphering framework, provable security
- M.Sc. in Mathematical Science Mar. 2016 - Feb. 2018
 - Where: KAIST, Daejeon, Korea
 - Advisor: Sanggeun Han
 - Research Area: cryptanalysis of LWE
- B.Sc. in Mathematics Mar. 2012 - Feb. 2016
 - Where: POSTECH, Pohang, Korea

Publications

Authors are listed in alphabetical order by last name, unless an asterisk(*) is indicated. Daggers (†) indicate co-first authors.

Academic Papers

1. **S. Kim**, B. Lee, and M. Son. “Shorter VOLE-in-the-Head-based Signatures from Vector Semi-Commitment”. The 33rd ACM Conference on Computer and Communications Security (**CCS 2026**), to appear.
2. W. Chung, S. Hwang, **S. Kim**, B. Lee, and J. Lee. “Security of NIST SP800-90A Hash-based DRBGs in the Seedless Model”. The 11th IEEE European Symposium on Security and Privacy (EuroS&P 2026).
3. W. Chung, S. Hwang, **S. Kim**, B. Lee, and J. Lee. “Making GCM Great Again: Toward Full Security and Longer Nonces”. The 44th Annual International Conference on the Theory and Applications of Cryptographic Techniques (**EUROCRYPT 2025**).
4. **S. Kim**, B. Lee, and M. Son. “Relaxed Vector Commitment for Shorter Signatures”. The 44th Annual International Conference on the Theory and Applications of Cryptographic Techniques (**EUROCRYPT 2025**).
5. K. Han, **S. Kim**, and Y. Son. “Private Computation on Common Fuzzy Records”. Proceedings on Privacy Enhancing Technologies Symposium (PoPETs 2025).
6. K. Han, **S. Kim**, B. Lee, and Y. Son. “Revisiting OKVS-based OPRF and PSI: Cryptanalysis and Better Construction”. The 30th Annual International Conference on the Theory and Application of Cryptology and Information Security (**ASIACRYPT 2024**).
7. ***S. Kim**[†], J. Ha[†], M. Son, B. Lee, D. Moon, J. Lee, S. Lee, J. Kwon, J. Cho, H. Yoon, and J. Lee. “AIM: Symmetric Primitive for Shorter Signatures with Stronger Security”. The 30th ACM Conference on Computer and Communications Security (**CCS 2023**).
8. J. Ha, **S. Kim**, B. Lee, J. Lee, and M. Son. “Rubato: Noisy Ciphers for Approximate Homomorphic Encryption”. The 41st Annual International Conference on the Theory and Applications of Cryptographic Techniques (**EUROCRYPT 2022**).
9. J. Cho, J. Ha, **S. Kim**, B. Lee, J. Lee, J. Lee, D. Moon, and H. Yoon. “Transciphering Framework for Approximate Homomorphic Encryption”. The 27th Annual International Conference on the Theory and Application of Cryptology and Information Security (**ASIACRYPT 2021**).
10. *J. Ha, **S. Kim**, W. Choi, J. Lee, D. Moon, H. Yoon, and J. Cho. “Masta: An HE-friendly Cipher Using Modular Arithmetic”. IEEE Access 10.1109/ACCESS.2020.3033564, 2020.
11. **S. Kim**, B. Lee, and J. Lee. “Tight Security Bounds for Double-Block Hash-then-Sum MACs”. The 39th Annual International Conference on the Theory and Applications of Cryptographic Techniques (**EUROCRYPT 2020**).

Preprints

- S. Kim, B. Lee, and M. Son. “Shorter VOLE-in-the-Head-based Signatures from Vector Semi-Commitment”. Cryptology ePrint Archive. Report 2025/1077. 2025. <https://eprint.iacr.org/2025/1077>.

Technical Report

- *J. Lee, J. Cho, J. Ha, **S. Kim**, J. Kwon, B. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMer Signature Scheme (Ver. 2.1)”. 2024. <https://aimer-signature.org>.

- *J. Lee, J. Cho, J. Ha, **S. Kim**, J. Kwon, B. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMER Signature Scheme (Ver. 2.0)”. Submission to Korean Post-Quantum Cryptography (KpqC) Competition 2nd Round. 2024. <https://aimer-signature.org>.
- ***S. Kim**, J. Ha, M. Son, and B. Lee. “Efficacy and Mitigation of the Cryptanalysis on AIM”. The 5th NIST PQC Standardization Conference. 2024. <https://eprint.iacr.org/2023/1474>.
- ***S. Kim**, J. Cho, M. Cho, J. Ha, J. Kwon, B. Lee, J. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMER Signature Scheme (Ver. 1.0)”. Submission to NIST Call for Additional Signature Schemes. 2023. <https://aimer-signature.org>.
- ***S. Kim**[†], J. Ha[†], M. Son, B. Lee, D. Moon, J. Lee, S. Lee, J. Kwon, J. Cho, H. Yoon, and J. Lee. “The AIMER Signature Scheme (Ver. 0.9)”. Submission to Korean Post-Quantum Cryptography (KpqC) Competition. 2022. <https://aimer-signature.org>.

Ph.D. Dissertation

- **S. Kim**. "On Homomorphic Encryption, Transciphering Frameworks, and HE-friendly Ciphers". 2022. KAIST.

Repositories

- <https://github.com/KAIST-CryptLab/RtF-Transciphering> RtF framework with HERA, Rubato
- <https://github.com/samsungsds-opensource/AIMer> AIMer

Talks and Posters

- “Hash-based Signature”. DGIST AI & Crypto Summer Camp 2026. Jul. 2026. Daegu, Korea.
- “The KpqC Competition”. Invited Talk, GlobalPlatform Security Task Force Meeting. Apr. 2026. Bangkok, Thailand.
- “The AIMER Signature Scheme”. KpqC Winter Camp 2026. Feb. 2026. Seoul, Korea.
- “Relaxed Vector Commitment for Shorter Signatures”. Invited Talk, ICISC 2025. Nov. 2025. Seoul, Korea.
- “AIMer Standard Format”. 2025 KpqC Conference. Nov. 2025. Seoul, Korea.
- “Hash-based Signatures”. KIISC Cryptography Education Program. Nov. 2025. Online.
- “AIMer Standardization Document Writing Plan”. 2025 KpqC Workshop. Jul. 2025. Taejeon, Korea.
- “Secure Multiparty Computation, MPC-in-the-Head, and AIMer”. KIISC Cryptography Education Program. Jul. 2025. Online.
- “AIMer v2.1 and Beyond”. Cryptography Research Society (in KIISC) 2025 Mid-term Workshop. Jun. 2025. Busan, Korea.
- “AIMer v2.1 and Beyond”. Invited Talk, KMS Spring Meeting 2025. Apr. 2025. Daejeon, Korea.
- “Revisiting OKVS-based OPRF and PSI: Cryptanalysis and Better Construction”. **Asiacrypt 2024**. Dec. 2024. Kolkata, India.

- “Updates on AIMer”. KpqC 9th Workshop. Oct. 2024. Seoul, Korea.
- “Circuit-PSI and Applications”. NIST Workshop on Privacy Enhancing Cryptography. Sep. 2024. Online.
- “AIMer”. KpqC Contest Colloquium. Aug. 2024. Seoul, Korea.
- (Poster) “The AIMer Signature Scheme”. NIST 5th PQC Standardization Conference. Apr. 2024. Rockville, MD, US.
<https://csrc.nist.gov/Events/2024/fifth-pqc-standardization-conference>.
- “Efficacy and Mitigation of the Cryptanalysis on AIM”. NIST 5th PQC Standardization Conference. Apr. 2024. Rockville, MD, US.
- “AIM: Symmetric Primitive for Shorter Signatures and Stronger Security”. **ACM CCS 2023**. Nov. 2023. Copenhagen, Denmark.
- “The AIMer Signature Scheme”. Invited Talk, 2nd Oxford PQC Summit. Sep. 2023. Oxford, United Kingdom. <https://www.maths.ox.ac.uk/events/conferences/past-events/oxford-post-quantum-cryptography-workshop-2023>.
- “Signature Schemes based on the MPC-in-the-Head Paradigm”. Invited Talk, Ewha-KMS International Cryptography Workshop 2023. Jul. 2023. Seoul, Korea.
- “Reducing the Overhead of Approximate Homomorphic Encryption”. Invited Talk, KMS Fall Meeting 2022. Oct. 2022. Seoul, Korea.
- “Rubato: Noisy Ciphers for Approximate Homomorphic Encryption”. **Eurocrypt 2022**. Jun. 2022. Trondheim, Norway. https://youtu.be/TE_sYzJtZQc (in English).
- “Transciphering Framework for Approximate Homomorphic Encryption”. Invited Talk, CryptoLab. Dec. 2021. Seoul, Korea.
- “Transciphering Framework for Approximate Homomorphic Encryption”. **Asiacrypt 2021**. Dec. 2021. Online. https://youtu.be/r3_07fWq0as (in English).
- “Transciphering Framework for Approximate Homomorphic Encryption”. Security @ KAIST. Nov. 2021. Online. <https://youtu.be/xKEgtZeMTaw?t=6434> (in Korean).
- “Hybrid Framework for Approximate Computation over Encrypted Data”. KMS Spring Meeting 2021. Apr. 2021. Online.

Patents

- K. Han, **S. Kim**, B. Lee, Y. Song, J. Seo, and I. Hwang. “Method, apparatus and computer program for bypassing verification in homomorphic encryption”. KR1020250193693. Dec. 2025.
- **S. Kim**, K. Han, and B. Lee. “Method for Multiple Applications of an Oblivious Pseudo-Random Function Protocol between a Receiving Device and a Sending Device based on an Oblivious Key-Value Store Algorithm, and Device using the Same”. KR1020250091679. Jul. 2025.
- B. Lee and **S. Kim**. “Method, Apparatus, System and Computer Program for Generating Variable-Output-Length Pseudo-Random Function based on Block Cipher”. KR1020250059321. May. 2025.
- **S. Kim**, B. Lee, M. Son. “Method, Apparatus, System and Computer Program for Zero-Knowledge Proof based on Binary Tree”. (KR1020250156598, US19189453, EP25172274). Feb. 2025.

- Y. Son, **S. Kim**, B. Lee, and K. Han. “Method for implementing a private set intersection protocol using an oblivious pseudorandom function based on minicrypt, and a terminal device using the same”. (KR1020250109577, US19005316). Jan. 2024.
- Y. Son, K. Han, **S. Kim**. “Method for Generating Common Identifier and Apparatus Therefor”. (KR1020240153894, US18396442, EP232189712). Sep. 2023.
- **S. Kim**, D. Moon, J. Kwon, S. Lee, J. Lee, M. Son, B. Lee, and J. Ha. “Method for Calculating using an One-Way Function Efficient in a Zero Knowledge Proof, and Apparatus Implementing the Same Method”. (KR1020240073510, US18387520, EP232031583). Nov. 2022.
- Y. Son, K. Han, **S. Kim**. “Method for Protecting Data Based on Private Set Union Protocol, and Apparatus Implementing the Same Method”. (KR1020240029490, US18216223, EP231846056). Oct. 2022.
- D. Moon, J. Lee, J. Lee, Y. Son, **S. Kim**, J. Ha, M. Son, and B. Lee. “Method for Calculating Using a Zero Knowledge Proof-Friendly One-Way Function, and Apparatus Implementing the Same Method”. (KR1020230161195, US18198667, EP231728221, SG10202301388U). May 2022.
- J. Lee, D. Moon, H. Yoon, J. Cho, **S. Kim**, J. Lee, and J. Ha. “Method and Apparatus for Generating Key Stream”. (KR1020220146115, US17514135). Apr. 2021.
- J. Lee, D. Moon, H. Yoon, J. Cho, E. Kim, **S. Kim**, J. Lee, J. Ha, W. Choi. “Apparatus and Method for Encryption, Apparatus and Method for Converting Ciphertext”. (KR20210129573, US17081862). Oct. 2020.
- D. Moon, H. Yoon, and J. Cho, **S. Kim**, J. Lee, J. Ha, and W. Choi. “Apparatus and Method for Symmetric-key Cryptography”. KR1020220022826. Oct. 2020.

Skills

- I speak Korean as a native and English fluently as a second language.
- Familiar with C/C++ (with x86 intrinsics), Python, Mathematica and $\text{\LaTeX}$
- Proficient to use PET libraries (e.g. SEAL, HELib, LibOTe)

Teaching Experiences

- Regular lecture (Korea University)
 - SCS503 Modern cryptography: Spring 2026
 - SMRT205 Mathematics for AI security: Spring 2026
- Counseling assistant (KAIST): Sep. 2016 - Feb. 2021
- Teaching assistant (KAIST)
 - IS511 Information security: Spring 2018, Spring 2019
 - CS204 Discrete mathematics: Fall 2019, Spring 2021

Services

Program committee:

- DPM Workshop 2026
- ISC 2025, 2026
- ICISC 2023, 2024, 2025

External reviewer:

- Eurocrypt 2023, 2026
- Asiacrypt 2019, 2020, 2024
- PKC 2024
- ICISC 2018
- ProvSec 2018

Honors and Awards

- The final selected digital signature algorithm (AIMer) in KpqC Competition (2025)
- The bronze award at Samsung Paper Award in 2023, 2025
- The best dissertation award at Korean Mathematical Society in 2023
- The 3rd award at iDash Competition (Track 4: Secure Record Linkage) in 2022
- The 2nd award at Korea Cryptography Contest (hosted by Korea Cryptography Forum) in 2018, 2024

Other Experiences

- Exchange student at NUS, Singapore in 2015
- Cellist in POSTECH orchestra (Mar. 2012 - Feb. 2016) / KAIST orchestra (Mar. 2016 - Feb. 2020)
- I have traveled to ...
 - Asia: JP, CN, TW, MY, ID, VN, SG, TH, LA, MM, IN, NP, MV, MN, KZ, IL
 - Europe: NO, CZ, PL, DE, NL, CH, IT, FR, ES, UK, DK, AT, VA, RU
 - America & Oceania & Africa: US (MD, NY, DC, GU), NZ, MA

Previous Email Addresses

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